

Final Exam Review

ECON 204/INTS 211: Contemporary Economic Issues

Part I: Income Distribution and Inequality

Question 1

Consider the following income distribution (in dollars):

5,000, 15,000, 25,000, 55,000, 100,000

- (a) Compute total income and mean income.
- (b) Compute the median income.
- (c) Compute the income share of the top 20% and bottom 40%.
- (d) Suppose the highest income increases to \$140,000. Recompute the mean and comment on inequality.
- (e) Explain why the mean may be misleading in skewed distributions.

Question 2 (Gini Coefficient)

Using the original income distribution:

- (a) Construct cumulative population shares and cumulative income shares.
- (b) Using the trapezoidal rule, compute the Gini coefficient:

$$G = 1 - \sum (Y_i + Y_{i-1})(X_i - X_{i-1})$$

- (c) Interpret the magnitude of the Gini coefficient.
- (d) Explain how a transfer from the richest to the poorest household would affect the Lorenz curve and Gini.

Part II: Inflation and Cost of Living

Question 3

A consumer price index evolves as follows:

$$CPI_{2024} = 132.5, \quad CPI_{2025} = 140.8$$

Nominal wages increase from \$52,000 to \$54,600.

- (a) Compute the inflation rate.
- (b) Compute nominal wage growth.
- (c) Compute real wage growth.
- (d) Determine whether purchasing power increased or decreased.
- (e) Explain how inflation can redistribute income between borrowers and lenders.

Question 4 (Household Inflation)

A household has the following expenditure shares:

Category	Weight	Inflation
Housing	45%	8%
Food	20%	6%
Transport	15%	4%
Other	20%	2%

- (a) Compute the household-specific inflation rate.
- (b) If official CPI inflation is 3.5%, explain the discrepancy.
- (c) Which households are most affected by this inflation pattern?

Part III: Monetary Policy and Interest Rates

Question 5

Suppose:

$$i = 6.5\%, \quad \pi^e = 3.2\%$$

- (a) Compute the real interest rate.
- (b) Explain how a rise in the policy rate affects:
 - (a) consumption
 - (b) investment
 - (c) exchange rates
- (c) Explain how monetary policy affects inflation expectations.

Part IV: Labour Markets and Inequality

Question 6

Wages for two groups evolve as follows:

Group	2005	2025
High-skill	\$28	\$70
Low-skill	\$20	\$25

- (a) Compute the wage gap in both years.
- (b) Compute percentage wage growth for each group.
- (c) Explain how skill-biased technological change explains the results.
- (d) Discuss one policy that could reduce inequality.

Part V: Housing Affordability

Question 7

A household earns \$95,000 annually and considers purchasing a home priced at \$855,000.

- (a) Compute the price-to-income ratio (PTI).
- (b) Interpret the result.
- (c) If interest rates increase, explain how affordability changes even if prices remain constant.

Question 8 (Mortgage Burden)

A household takes a mortgage of \$600,000.

- (a) Explain qualitatively how monthly payments change when interest rates rise.
- (b) Define and explain a **mortgage renewal shock**.
- (c) Explain why this is particularly important in Canada.

Question 9 (Rental Market)

Monthly income = \$4,200.

Rent increases from \$1,300 to \$2,050.

- (a) Compute rent-to-income ratios before and after.
- (b) Explain affordability thresholds.
- (c) Discuss how rising rents contribute to inequality.

Part VI: Integrated Policy Analysis

Question 10

Consider a policy that increases housing supply through zoning reform.

- (a) Explain how this affects housing prices in the short run and long run.
- (b) Discuss the effect on inequality.
- (c) Identify one unintended consequence.

Question 11 (Integrated Analysis)

Suppose inflation rises to 6% and the central bank increases interest rates significantly.

- (a) Explain the impact on:
 - (a) housing demand
 - (b) mortgage payments
 - (c) rental markets
- (b) Explain why inflation may fall but housing affordability may worsen.
- (c) Discuss distributional effects across income groups.

End of Review